FluxFormer: A Transformer Surrogate for Differentiable Genome-Scale Metabolic Modeling

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ABSTRACT

Genome-scale metabolic models (GEMs) can be used to predict metabolic phenotypes, but making accurate quantitative predictions from experimental data is still challenging. In flux balance analysis (FBA), the experimental medium composition must be translated into the uptake constraints required by the model[1]. Multi-omics measurements are also difficult to integrate directly into GEM-based predictions[2]. One approach to using such experimental data is the reservoir configuration introduced in the Artificial Metabolic Network (AMN) and later adopted in the Metabolic-Informed Neural Network (MINN): a differentiable approximation of FBA is pretrained on simulated data, frozen, and coupled to a neural network that infers its exchange constraints from experimental inputs[2,3]. We develop FluxFormer, a Transformer-based FBA-surrogate, and study how the choice of simulated pretraining data affect its performance on experimental tasks. FluxFormer represents each metabolic reaction as a token, and self-attention determines how information is exchanged between reaction tokens. We train it on parsimonious FBA (pFBA) simulations of the full, unreduced 2,712-reaction *Escherichia coli* GEM iML1515[4] and predict the complete steady-state flux distribution without reaction-set reduction. We generate simulated pretraining data from six different distributions. Two are based on the AMN and MINN settings, while the others combine these settings or extend them to a broader range of nutrient conditions. The broader distributions use a literature-informed pool of physiologically relevant organic carbon sources. A separate FluxFormer is trained on each distribution. We first test each pretrained model on held-out pFBA simulations to measure how accurately they reproduce the solver. We then use the frozen models for two experimental tasks. For growth prediction from medium composition, a prior network coupled to FluxFormer reaches $R^2 = 0.88$, exceeding the published AMN results[3]. In the MINN-style multi-omics task[2], the inferred exchange constraints improve standard pFBA predictions. The learned internal representations for reaction tokens form distinct reaction-specific clusters, and their positions shift with nutrient composition. The results show that FluxFormer can transfer from simulated pretraining to experimental prediction even when it is trained on a broad nutrient space rather than conditions tailored to a specific downstream task. Using *E. coli* as the first genome-scale example, this is a step toward foundation models for microbial metabolism.

Introduction

Genome-scale metabolic models (GEMs) encode the biochemical reaction network inferred from an organism's genome. Constraint-based modeling uses these reconstructions together with physicochemical and environmental constraints to characterize feasible metabolic states and predict cellular phenotypes[5]. Flux balance analysis (FBA) is one of the most widely used constraint-based methods[6,7]. By assuming a metabolic steady state and optimizing an objective such as biomass production subject to flux bounds and stoichiometric constraints, FBA maps a genome-scale reconstruction to a phenotype predictor. For *Escherichia coli*, iML1515 provide a well-characterized genome-scale reconstruction for predicting metabolic traits under different conditions[4]. One challenge for quantitative FBA prediction is specifying uptake constraints: FBA requires condition-specific bounds on nutrient uptake reactions, but these bounds are not directly determined by the experimentally controlled medium[7]. These constraints are often approximated from available measurements or assumed values, and their specification can strongly affect downstream predictions[1,8].

Hybrid methods that combine machine learning with constraint-based metabolic models have been developed in several forms[9–11]. For example, Culley et al. combined pFBA-derived metabolic fluxes with gene-expression data in a neural network to predict yeast growth[12]. Other approaches have combined FBA with regularized machine-learning models[13], graph neural networks[14], or neural networks that predict flux constraints for GEMs[15]. Another design uses a differentiable surrogate of FBA that is first pretrained on simulated flux data and then frozen. A lightweight prior network is subsequently trained on experimental data to infer exchange constraints from inputs such as medium composition. This configuration was introduced as

the *reservoir* approach in the Artificial Metabolic Network (AMN)[3] and was later adopted in the Metabolic-Informed Neural Network (MINN)[2]. The two implementations differ in the surrogate they use and the experimental inputs they accept.

Neural networks typically require large training datasets, which is a particular problem in biology where experimental data is often limited and costly to obtain[16]. Simulated data provide a way to pretrain a model before it is applied to a smaller experimental dataset. This idea is also used outside metabolic modeling. Tabular foundation models such as TabPFN and TabICL are pretrained on large collections of synthetic tabular datasets before being applied to unseen real-world datasets[17,18]. In protein modeling, METL pretrains Transformer models on biophysical simulation data before fine-tuning them on experimental protein measurements[19]. For metabolic modeling, FBA can generate large amounts of structured training data without requiring additional experiments. The simulated conditions used to generate data are therefore important, since they determine which metabolic responses the surrogate learns during pretraining.

In the reservoir approach, the surrogate is central because the prior network is trained through it. Its accuracy therefore affects the training signal the prior network receives. Existing reservoir implementations used reduced reaction representations in their genome-scale experiments. In AMN, reactions that remained inactive across the simulated training conditions were removed, reducing the iML1515 representation to 550 reactions. MINN selected the smaller iAF1260 model to lower the computational cost of training and further evaluated reduced variants that excluded reactions unable to carry flux under the relevant conditions[2,3]. In contrast, we train the surrogate developed in this work on pFBA simulations generated from the full 2,712-reaction iML1515 model.

We propose **FluxFormer**, a Transformer[20] surrogate for the reservoir approach. FluxFormer treats metabolic reactions as tokens and couples them through attention. Trained on pFBA simulations of the full, unreduced 2,712-reaction iML1515 model, it predicts the complete steady-state flux distribution without reaction-set reduction. We also study how the simulated pretraining distribution affects transfer to experimental tasks. Previous reservoir approaches used simulated conditions designed around their downstream experimental settings[2,3]. In this work, we also train on broader nutrient spaces to see whether our surrogate still transfers well to specific experimental tasks. We use *E. coli* as the test case, with the following contributions:

- **Simulated pretraining across different nutrient spaces**. We generate task-specific, combined, and broader pretraining distributions and evaluate how the simulated condition space affects surrogate fidelity and transfer to experimental data.
- **The same surrogate architecture across both downstream tasks.** We use FluxFormer as the frozen metabolic surrogate in both experiments. For growth prediction, the prior network coupled to FluxFormer reaches $R^2 = 0.88$, above the published AMN-QP, AMN-LP, and AMN-Wt results[3]. In the MINN-style missing-flux experiment, the exchange constraints inferred through FluxFormer improve the standard pFBA prediction.
- **Reaction representations reflect nutrient conditions.** The learned embeddings form distinct clusters for individual reactions, and their positions within each cluster shift with nutrient composition.

The immediate goal is to determine whether a Transformer pretrained on simulated pFBA data can provide a useful metabolic prior for experimental prediction and how it compares with the existing AMN and MINN approaches. We test this through growth prediction from medium composition and by inferring missing exchange constraints that improve pFBA predictions from multi-omics data. We also examine how these results change as the simulated pretraining space becomes broader. Automated reconstruction has expanded the availability of GEMs across microbial species[21]. The longer-term goal is to develop a single pretrained metabolic surrogate that can be reused across nutrient conditions and ultimately applied across multiple organisms.

Background: The Reservoir Approach to Hybrid Metabolic Modeling

Flux Balance Analysis

FBA models metabolism as a stoichiometric network. For a GEM with m metabolites and n reactions, a stoichiometric matrix $S \in \mathbb{R}^{m \times n}$ relates reactions to the metabolites they consume and produce. FBA assumes a steady state, where the flux vector $v \in \mathbb{R}^n$ satisfies the condition $Sv = 0$ subject to lower and upper bounds given by $v_{\text{lower}} \leq v \leq v_{\text{upper}}$. These bounds define the allowable range of each reaction and, for exchange reactions, determine which nutrients can be taken up from the medium. FBA selects among feasible fluxes by maximizing a linear objective $c^\top v$, typically biomass production[22], yielding a linear programming problem[6]. Parsimonious FBA (pFBA) first maximizes the primary objective and then minimizes the total absolute flux while maintaining that optimum[23].

A major source of uncertainty in quantitative FBA predictions is specifying uptake bounds. The exchange-reaction bounds that represent a medium are key model inputs, and mapping an experimentally defined medium to the uptake constraints is not captured by stoichiometry alone[1]. Reservoir approaches address this problem by learning exchange constraints from experimental inputs[2,3].

The Reservoir Approach

The reservoir approach is a two-stage scheme that combines experimental data with a frozen differentiable surrogate of FBA[2,3].

Stage 1 – surrogate pretraining.

A differentiable model f_θ is trained on simulated pFBA solutions to approximate the mapping from exchange-reaction constraints to the resulting steady-state flux distribution, $V_{\text{in}} \mapsto V_{\text{out}}$. The simulations provide large labeled training datasets without additional experimental measurements[24]. After pretraining, its parameter θ are fixed.

Stage 2 – prior-network fitting.

A prior network g_ϕ is trained on experimental data to map the available measurements C to exchange constraints V_{in}. The frozen surrogate produces $f_\theta(g_\phi(C))$, which can be compared with an experimental target such as growth rate or measured fluxes. Gradients pass through the surrogate during training, while only the parameters ϕ of the prior network are updated.

The downstream model therefore depends on both the accuracy of the frozen surrogate and the simulated conditions on which it was pretrained. These two factors are central to the experiments in this work.

Existing surrogates: AMN and MINN

AMN introduced the reservoir configuration used here[3]. The AMN is first trained on FBA simulations and then frozen. A separate prior network learns to predict uptake bounds from the experimental medium composition. Faure et al. evaluated this configuration on *E. coli* growth-rate regression and *Pseudomonas putida* growth classification. For the genome-scale *E. coli* experiments, reactions that remained inactive across the simulated training conditions were removed, giving a reduced iML1515 representation with 550 fluxes. A dense Artificial Neural Network (ANN) trained on this representation performed poorly and had not converged after 100 epochs[3]. More recently, dAMN extended the framework to dynamic bacterial growth modeling[25].

MINN later extended the reservoir approach to multi-omics data[2]. Its pretrained block maps selected exchange constraints to a complete flux distribution, while a prior network infers missing exchange constraints from omics measurements and available exchange fluxes. The inferred constraints are then supplied to standard pFBA. MINN selected the smaller iAF1260 reconstruction instead of iML1515 to reduce training cost and also evaluated FVA- and FBA-reduced representations containing 1,873 and 587 split reactions, respectively[2,26].

FluxFormer as a surrogate

FluxFormer serves the same role as the frozen surrogate f_θ in the reservoir framework. It represents metabolic reactions as tokens and couples them through attention. Unlike previous setups, FluxFormer is pretrained on pFBA simulations of the full, unreduced 2,712-reaction iML1515 reconstruction[4]. The same pretrained architecture is then used in both the AMN-style growth-prediction task and the MINN-style missing-flux task. The Architecture section describes the model in detail.

Methods

Overview of the Study

Figure 1 summarizes the overall workflow used in this study. We first pretrain FluxFormer as a surrogate for pFBA using simulated data generated from the full iML1515 model. For each simulated nutrient condition, we sample a set of exchange-reaction bounds and use pFBA to compute the corresponding steady-state flux distribution. Each sampled set of bounds and the resulting flux distribution form one training example for FluxFormer. This follows our previous work[27]. Because we generate these samples in silico before training, we can scale the dataset as needed. In this study, we pretrained each model on one million samples.

After pretraining, we first evaluate surrogate performance on independently generated simulated conditions. We then freeze the pretrained FluxFormer and transfer it to two experimental prediction tasks. In each task, we train a small multilayer perceptron (MLP) in front of the frozen surrogate to map the available experimental measurements to the exchange constraints FluxFormer requires. During training, gradients propagate through FluxFormer, but only the MLP parameters are updated. We repeat the same procedure for FluxFormer models pretrained with different simulated condition distributions.

Architecture

The left panel of Figure 2 shows a single FluxFormer layer. FluxFormer represents each reaction as a token and maintains two aligned representations throughout the network: a learned embedding for each reaction and a scalar channel carrying its flux value. The embeddings determine how reactions interact through self-attention, and the resulting attention weights diffuse the scalar values across reactions. We describe the architecture in more detail in our previous work[27]. The architecture is motivated by the coupling between reaction fluxes imposed by stoichiometric mass balance and reaction constraints in FBA[8,28]. Related architectures also maintain coupled representation streams, including graph Transformers, AlphaFold's Evoformer, and equivariant molecular Transformers[29–31].

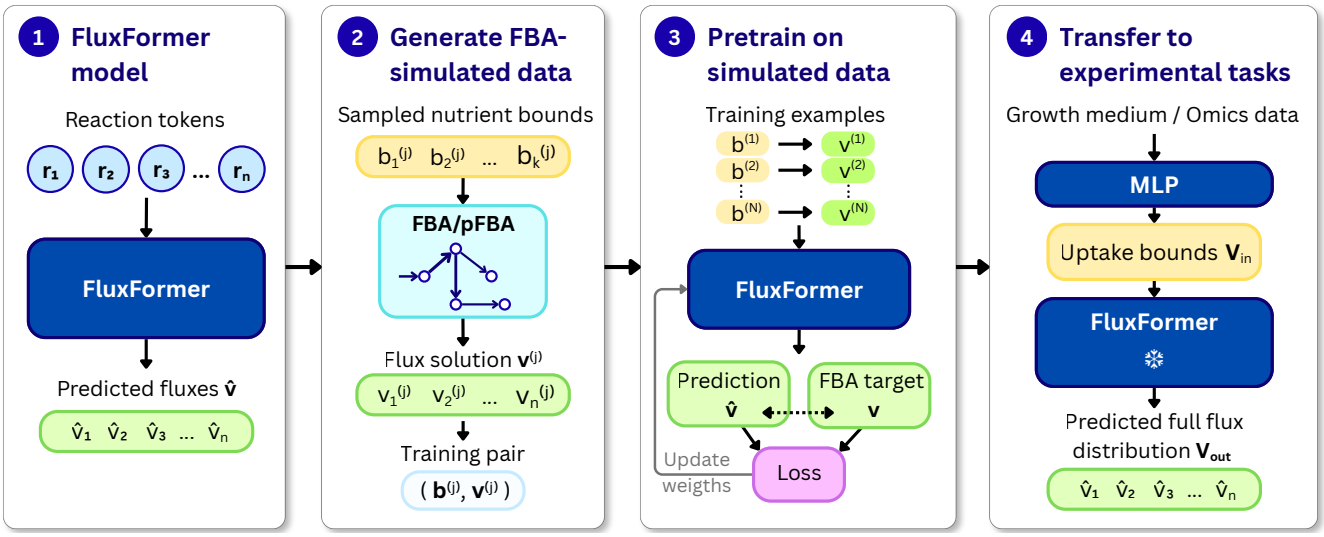


Figure 1. Overview of the pretraining of FluxFormer and its application to experimental tasks.

The reconstruction contains n reactions, each represented by one token; for iML1515, $n = 2{,}712$[4]. Reaction i has a learned embedding $x_i \in \mathbb{R}^d$ and a scalar state $c_i \in \mathbb{R}$. At the input, c_i is set to the sampled exchange constraint for reactions constrained by the medium and to zero for all other reactions. Collecting all reaction tokens gives $X \in \mathbb{R}^{n\times d}$ and $c \in \mathbb{R}^n$.

Each FluxFormer layer updates the reaction embeddings and scalar states. The attention block first applies pre-layer normalization and multi-head self-attention to the embeddings[20,32]. The attention matrix from each head is kept separately and later used to update the scalar channel. Writing $A^{(h)} \in \mathbb{R}^{n\times n}$ for the attention matrix of head h, the embeddings follow the standard residual attention update, $X \leftarrow X + \mathrm{MHA}(\mathrm{LN}(X))$. The scalar channel is then diffused through the same attention matrices. Each head updates c according to its attention distribution, $c^{(h)} = A^{(h)}c$. Let $s = (s_1, \ldots, s_H) \in \mathbb{R}^H$ contain one learned score for each of the H attention heads, and define the normalized head weights as $\alpha = \mathrm{softmax}(s) \in \mathbb{R}^H$. The head-specific scalar updates are then combined as

$$c \leftarrow c + \sum_{h=1}^{H} \alpha_h A^{(h)} c.$$

The residual term carries the current scalar state forward while the weighted attention update adds information from other reactions. We do not use positional encoding because reaction order has no meaning, consistent with attention-based models for unordered sets[33].

The feed-forward block then processes the two representations jointly. The scalar channel is concatenated with the embeddings to form a $(d+1)$-dimensional vector, which passes through a pre-norm two-layer position-wise network with GELU activation[34] and a residual connection. The concatenated output is split back into an updated embedding of width d and an updated scalar, keeping the coupling representation and the flux representation linked through both sub-blocks of the layer.

Stacking L such layers, the model takes the initial embeddings and injected scalar channel and produces a scalar value for each reaction token at every layer. The prediction for reaction i is taken directly from its scalar value at the final layer, while the intermediate values are retained for analysis. The embeddings determine how reactions couple, while the scalar state is diffused and refined as it passes through the layers. The reported model uses $d = 256$, $H = 8$ heads, $L = 4$ layers, and a feed-forward width of 1,024, giving approximately 3.9M parameters.

Simulated pretraining data

We generated simulated nutrient conditions using the full iML1515 reconstruction and computed a parsimonious flux distribution for each condition, with biomass production as the objective. Each simulated sample consists of a set of exchange-reaction bounds describing the nutrient condition and the corresponding complete pFBA flux solution. Algorithm 1 summarizes the common generation procedure. The pretraining distributions used in the experiments are described in the Experiments and Results section.

All simulated conditions shared the same fixed basal exchanges, corresponding to the M9-based background used in the AMN study. These basal exchanges were applied before sampling the distribution-specific nutrients. Uptake of other organic

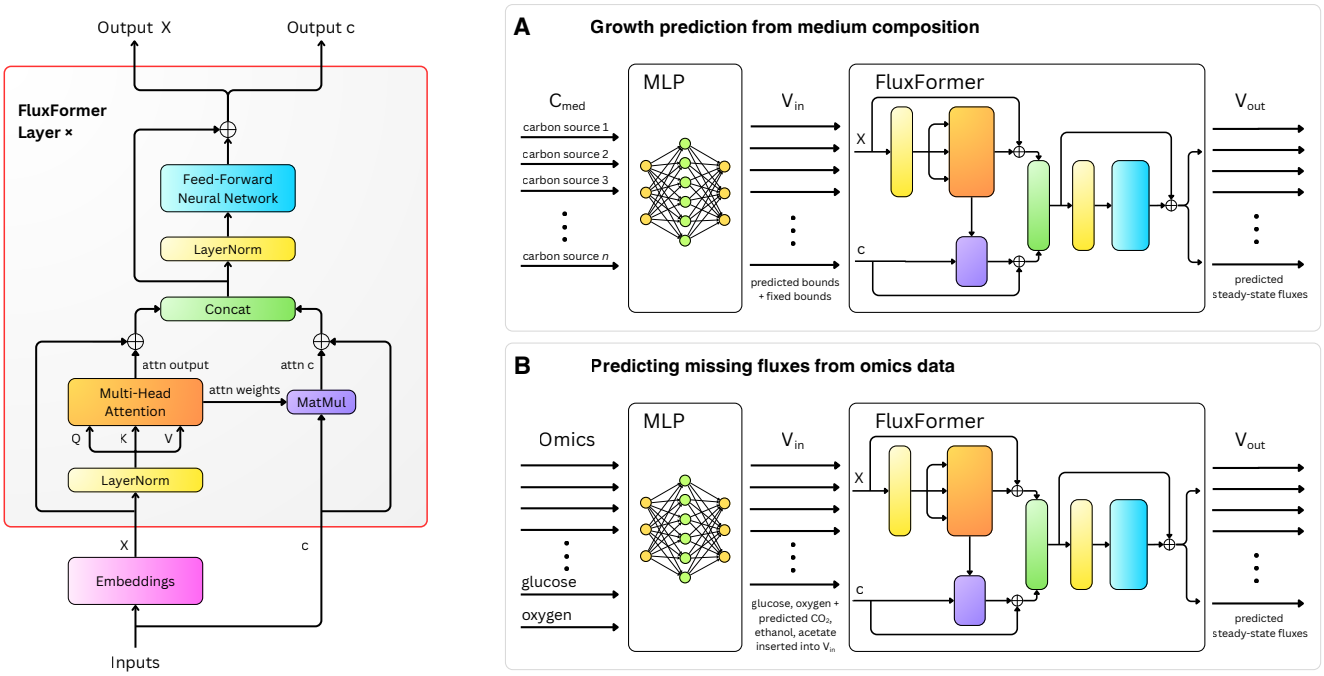


Figure 2. FluxFormer architecture and experimental prediction tasks. Left: a single FluxFormer layer updates reaction embeddings X and scalar flux values c through multi-head attention and a feed-forward block. Right: experimental pipelines for growth prediction and missing-flux prediction using a pretrained FluxFormer.

sources was initially closed, after which nutrients and their exchange bounds were sampled according to the selected pretraining distribution.

The resulting model was solved by pFBA. Conditions for which a valid solution could not be obtained were rejected and resampled. The sampled exchange limits were stored in a fixed input order as the surrogate input, while the complete reaction-flux vector was stored as the prediction target. Exact exchange identities, bound ranges, and other sampling parameters are given in the Supplementary.

Pretrain on simulated data

For each distribution, we trained a separate FluxFormer using one million input–flux pairs. We inserted the sampled exchange bounds into $\mathbf{x}^{(j)}$ at their corresponding reaction-token positions, and used all 2,712 pFBA-computed reaction fluxes in $\mathbf{v}^{(j)}$ as targets. Within each dataset, we kept the input-to-token mapping and reaction order fixed.

We split each dataset into an 80% training set and 20% validation set, using the validation set to choose the checkpoint. The models had the same embedding dimension of 256, eight attention heads, four Transformer layers, and a dropout rate of 0.02. During training, we minimized the mean Huber loss over the entire reaction vector by using AdamW with a learning rate of 10^{-4}. We monitored loss on the held-out split and retained the weights from the best epoch. We then fixed these weights for the later experimental tasks.

Transfer to experimental tasks

After pretraining, FluxFormer can serve as a differentiable surrogate for experimental prediction tasks. A task-specific multilayer perceptron (MLP) is placed in front of the frozen surrogate and maps the available experimental variables to the exchange constraints required by FluxFormer. These inputs can depend on the experimental setting and may include, for example, medium composition, omics measurements, measured exchange fluxes, or environmental variables such as pH and temperature. The MLP therefore links experimental observations to the exchange constraints used by the metabolic surrogate.

During downstream training, the predicted exchange bounds are passed to FluxFormer, which them to the corresponding flux distribution. Gradients propagate through the frozen FluxFormer, while only the parameters of the front MLP are updated. This allows the same pretrained surrogate to be used with different types of experimental data without retraining FluxFormer itself.

The right panel of Figure 2 shows the two applications considered in this study. In the growth-prediction task, the MLP

Algorithm 1 Generation of simulated input–flux pairs

Require: iML1515 reconstruction, sampling distribution d, accepted-sample count N

for $j = 1, \ldots, N$ **do**
  repeat
    Reset exchange bounds and close uptake of non-basal organic sources
    Apply the common basal exchange bounds
    Determine the sampling regime specified by distribution d
    Select variable nutrients according to the selected regime
    Sample their exchange bounds according to the selected regime
    Solve by pFBA at 0.999 of the optimal biomass value
  until a valid pFBA solution is obtained
  Store the sampled exchange limits in a fixed input order as $\mathbf{x}^{(j)}$
  Store the complete reaction-flux solution as $\mathbf{v}^{(j)}$
end for

maps medium composition to uptake bounds, and FluxFormer predicts biomass flux. In the missing-flux task, the MLP maps omics measurements and known exchange fluxes to the missing exchange constraints, which are then used to predict fluxes.

Experimental datasets

Growth prediction from medium composition

The growth-prediction experiment used the experimental *E. coli* dataset reported by Faure et al.[3]. The dataset contains 110 M9-based growth conditions with L-alanine, L-proline, L-threonine, and glycine present in all media, while ten carbon sources were varied between conditions. The carbon sources were represented as binary variables indicating whether each compound was present. The dataset contains 10 media with one variable carbon source, 20 with two, 40 with three, and 40 with four. Because the four amino acids were fixed across all conditions, the measured growth rates represent the response to the variable carbon-source composition against the same M9 background containing these amino acids. Each condition was measured in multiple technical replicates, and the mean maximum growth rate across replicates was used as the prediction target.

Missing-flux prediction from omics data

The missing-flux experiment followed the MINN setting based on the Ishii et al. *E. coli* dataset[2,35]. The dataset contains 29 glucose-limited chemostat conditions: five wild-type samples measured at different dilution rates and 24 single-gene knockout samples measured at the reference dilution rate. For each condition, the data include transcriptomic measurements for 79 genes, proteomic measurements for 60 proteins, and measured values for 47 metabolic fluxes. The evaluation used the FBA-fitted fluxomics target data from the MINN study, allowing direct comparison with the reported MINN results.

The FBA-fitted fluxomics data used by MINN should be interpreted with caution. The original Ishii fluxes do not lie within the feasible solution space of the genome-scale model iAF1260. Tazza et al. therefore projected each flux vector into the FBA solution space by finding the feasible vector closest to the original measurements in Euclidean distance[2]. This fitting minimizes the distance over the complete flux vector and does not necessarily preserve individual experimentally derived fluxes. In some cases, the resulting changes are substantial. For example, for WT_0.1h-1 the magnitude of the glucose exchange flux (EX_glc__D) changes from 1.34 in the original data to 5.862 in the FBA-fitted data, more than a fourfold increase. Thus, the fitted fluxomics data represent model-adjusted targets that are feasible under the selected GEM rather than the original experimental measurements. Evaluation therefore reflects agreement with this FBA-based projection.

We evaluated the 29 experimental conditions using leave-one-out cross-validation, comparing the predicted ad reference values of the 47 fluxes in each held.out condition. We report squared Pearson correlation (r^2), as calculated in the released implementations of Gonçalves et al. and Tazza et al.[2,36]. This differs from the conventional coefficient of determination (R^2) defined in the published methodology. We also calculated MAE, RMSE, and normalized error (NE), which measures Euclidian prediction error relative to the norm of the reference flux vector. Results are reported as means and population standard deviatins across the 29 conditions.

Experiments and Results

Pretraining distributions

The main experimental variable was the distribution from which the nutrient conditions were sampled. We considered six pretraining distributions, denoted A, B, A ∪ B, C, D, and E. Figure 3 shows how they relate to the AMN and MINN condition spaces. A and B were based on the AMN and MINN settings, respectively. A ∪ B combined conditions from these two spaces,

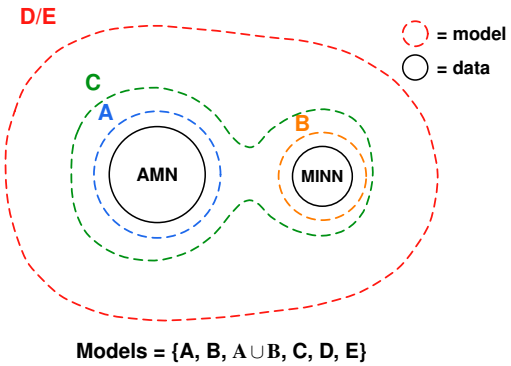


Figure 3. Conceptual relation between the AMN and MINN experimental condition spaces and the simulated pretraining distributions. A and B are task-specific distributions, A ∪ B is their literal mixture, C additionally covers intermediate task-relevant conditions, and D and E share the broader condition space but differ in how sampling probability is allocated within it.

C also included conditions between them, and D and E extended pretraining to a much broader set of organic carbon sources and nutrient combinations.

Distribution A was based on the AMN nutrient setting, but expanded its simulated condition space. It used the same ten variable carbon sources and retained glycerol and four supplemented amino acids as fixed inputs. Unlike the original AMN reservoir, which generated simulations from the 110 carbon-source combinations present in the experimental dataset, A sampled one to four of the ten sources independently. Oxygen was also varied in A rather than kept fixed as in the AMN simulations.

Distribution B followed the glucose-minimal conditions used by MINN, with glucose as the only supplied carbon source and oxygen and selected secretion bounds varied. A ∪ B contained equal numbers of independently sampled A- and B-style conditions without combining their carbon sources. Distribution C sampled A-like and B-like conditions with probabilities of 0.40 each, while the remaining 0.20 consisted of mixed conditions formed by adding up to three A carbon sources to a glucose-based B condition.

Distributions D and E used a common pool of 48 organic carbon sources, including those represented in the A and B condition spaces. The pool was constructed by first screening organic exchanges in iML1515 for their ability to support growth and then using an LLM-assisted literature review to identify physiologically relevant compounds reported in industrial and agricultural side streams.

For each general D or E condition, between one and ten distinct sources were drawn from this pool. The upper limit of ten was chosen to include complex mixtures while keeping the number of possible source combinations manageable. The number of selected sources followed a truncated geometric distribution, with an average of approximately three sources per condition. This gave more weight to smaller mixtures, whose combination space can be sampled more densely. Larger mixtures were still included, but less frequently. Oxygen was varied independently. Distribution E consisted entirely of these general conditions. Distribution D used the same broad space for 80% of its samples, with the remaining 10% drawn from A-style conditions and 10% from B-style conditions. This increased the sampling density around the two task-specific condition spaces.

Detailed definitions of all distributions, including the exact exchange identities, sampling probabilities, exchange-bound ranges, and D/E source-pool curation procedure, are given in the Supplementary.

Simulated surrogate fidelity

We first evaluated how accurately each pretrained FluxFormer reproduced simulated fluxes within its pretraining distribution. Each model received medium constraints from an independently generated test dataset, and its predictions were compared with the corresponding simulated reference fluxes. Inference used the complete vocabulary of 2,712 output fluxes. We computed pooled R^2, mean absolute error (MAE), and root mean squared error (RMSE) over all test conditions and predicted fluxes.

All six models reproduced their simulated reference fluxes with high accuracy as shown in Table 1. Pooled R^2 values were very close to 1 for all models, with low MAE and RMSE across all pretraining distributions. Model B gave the lowest errors, consistent with its small set of variable inputs and narrow condition space. Model E had the highest errors, but it was trained on the broadest nutrient distribution and still achieved $R^2 = 0.9980$. This shows that FluxFormer can closely approximate pFBA even over a highly general nutrient space.

Table 1. Simulated surrogate fidelity on independently generated test data from each model's pretraining distribution. Metrics are pooled across test conditions and all 2,712 output fluxes.

Model	R^2	MAE	RMSE
A	0.9997	0.0032	0.0283
B	1.0000	0.0020	0.0090
A ∪ B	0.9998	0.0040	0.0186
C	0.9998	0.0037	0.0207
D	0.9993	0.0033	0.0246
E	0.9980	0.0064	0.0614

Table 2. Growth-prediction performance on 110 experimental media, pooled over three repeats of stratified 10-fold cross-validation.

Model	R^2	MAE (h^{-1})	RMSE (h^{-1})
A	0.8735	0.0233	0.0300
A ∪ B	0.8747	0.0235	0.0298
C	0.8723	0.0239	0.0301
D	0.8669	0.0240	0.0308
E	0.8611	0.0244	0.0314
TabPFN	0.8063	0.0303	0.0371

Growth prediction from medium composition

We compared the pretrained surrogates on the experimental growth-prediction task to examine how the choice of pretraining distribution affects performance. For each model, we trained a separate front MLP while keeping FluxFormer frozen. We also tested TabPFN-3, which predicted growth rates directly from medium composition. The purpose was to compare FluxFormer with a model that had no prior training on metabolic simulations. All models were evaluated using the same stratified 10-fold cross validation splits and evaluation was repeated three times. We excluded model B because its training focused on MINN-style exchange constraints and did not include any of the variable carbon sources needed for the AMN growth-prediction task. Table 2 summarizes the pooled out-of-fold results.

All five FluxFormer models achieved similar accuracy despite their different pretraining distributions. Models A, A ∪ B, and C performed almost identically. Models D and E had slightly lower accuracy, but the differences were small considering their substantially broader pretraining conditions. In particular, model E had nearly the same predictive accuracy as the AMN-specific model A, despite being pretrained without a dedicated AMN sampling regime. This relatively small difference matters when developing more general metabolic surrogates.

All FluxFormer models outperformed TabPFN-3 and achieved higher prediction accuracy than the AMN models reported by Faure et al.[3] ($Q^2 \approx 0.78$). Faure et al. use Q^2 to denote the cross-validated coefficient of determination which is equivalent to the held-out R^2 used here. Figure 4 compares predicted and measured growth rates for models A and E, showing similar performance despite different pretraining distributions

Missing-flux prediction from omics data

We evaluated the front MLP coupled to frozen model B by comparing its final constrained pFBA predictions with the iML1515 pFBA baseline (Table 3). With the available glucose and oxygen inputs retained, adding the inferred carbon dioxide, ethanol, and acetate secretion caps increased mean r^2 from 0.893 to 0.908 and reduced RMSE from 0.832 to 0.774. MAE and normalized error also decreased.

The results are close to those reported by Tazza et al.[2] whose iAF1260-based MINN-Reservoir + pFBA achieved a reported score of 0.910. This demonstrates that FluxFormer achieves comparable performance using the full iML1515 reconstruction.

We limited the main comparison to model B whose pretraining conditions were designed for the MINN-style task. With only 29 conditions, sensitivity to target fitting and influential individual samples complicated interpretation of differences between pretraining distributions.

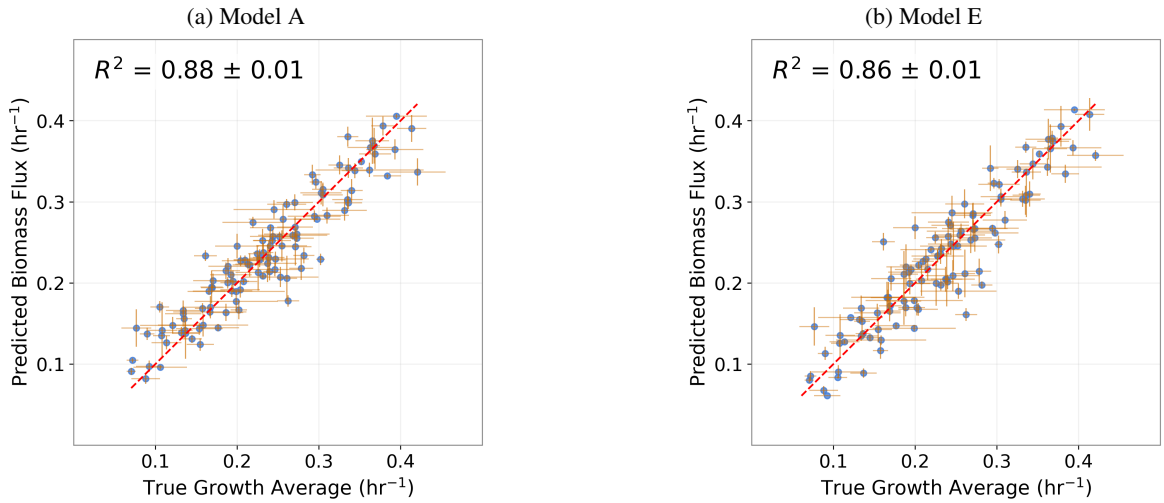


Figure 4. Predicted versus measured growth rates for FluxFormer models A and E.

Table 3. Performance on the missing-flux prediction task. The first two rows report published results from Tazza et al.[2]. Squared Pearson correlation (r^2) follows the released MINN implementation, although the original paper reports this metric as R^2.

Model	r^2	MAE	RMSE	NE
pFBA (iAF1260-based)	0.892 ± 0.127	0.496 ± 0.353	0.836 ± 0.625	0.306 ± 0.367
MINN-Reservoir + pFBA	0.910 ± 0.091	0.445 ± 0.261	0.740 ± 0.421	0.253 ± 0.159
pFBA (iML1515)	0.893 ± 0.132	0.495 ± 0.366	0.832 ± 0.634	0.309 ± 0.373
Model B + pFBA (iML1515)	0.908 ± 0.111	0.455 ± 0.322	0.774 ± 0.546	0.284 ± 0.312

Nutrient-dependent reaction representations

We examined how model E represented nutrient availability using 4,000 independently sampled test media and a controlled sweep of 10,000 input conditions. For the sweep, fructose and oxygen uptake bounds were varied independently over a 100×100 grid, while the all other medium constraints were held fixed. Each condition was passed through the pretrained FluxFormer with its weights frozen. We selected glycolysis as a representative pathway and extracted the 256-dimensional representations of 14 glycolysis-related reactions after the final Transformer layer. We then used t-SNE[37] to construct separate projections for the sampled media and the controlled sweep.

Figure 5 shows that different reactions occupy clearly separate regions. For the sampled test media (left), conditions were ordered by repeatedly selecting the nearest remaining medium, using Euclidian distance between standardized input vectors. Colors follow this ordering, with each condition assigned the same color across reactions. Similar color patterns recur across reactions regions, indicating that the representations vary systematically with medium composition. The controlled sweep (middle and right) reveals this dependence along two specified input variables. Both panels show the same coordinates, with the middle panel colored by fructose uptake bounds and the right by oxygen uptake bounds. The resulting gradients differ in shape and direction across the reaction regions.

Each reaction starts from a learned token embedding that is identical across input conditions. As FluxFormer processes the supplied medium, attention propagates information from the exchange-reaction inputs to other reaction tokens. Variation within each reaction region therefore arises during the forward pass and reflects how representations of internal glycolytic reactions respond to changing nutrient constraints.

These observations show that the final representations combine reaction identity with information about the current medium. The sampled conditions reveal this organization across a broad nutrient distribution, while the controlled sweep shows how the internal states change systematically with specified nutrient constraints.

Discussion

FluxFormer outperformed the published AMN models[3] in terms of growth prediction accuracy. This was achieved using the full 2,712-reaction iML1515 model, whereas the AMN models used a reduced set of reactions. Keeping all reactions is especially

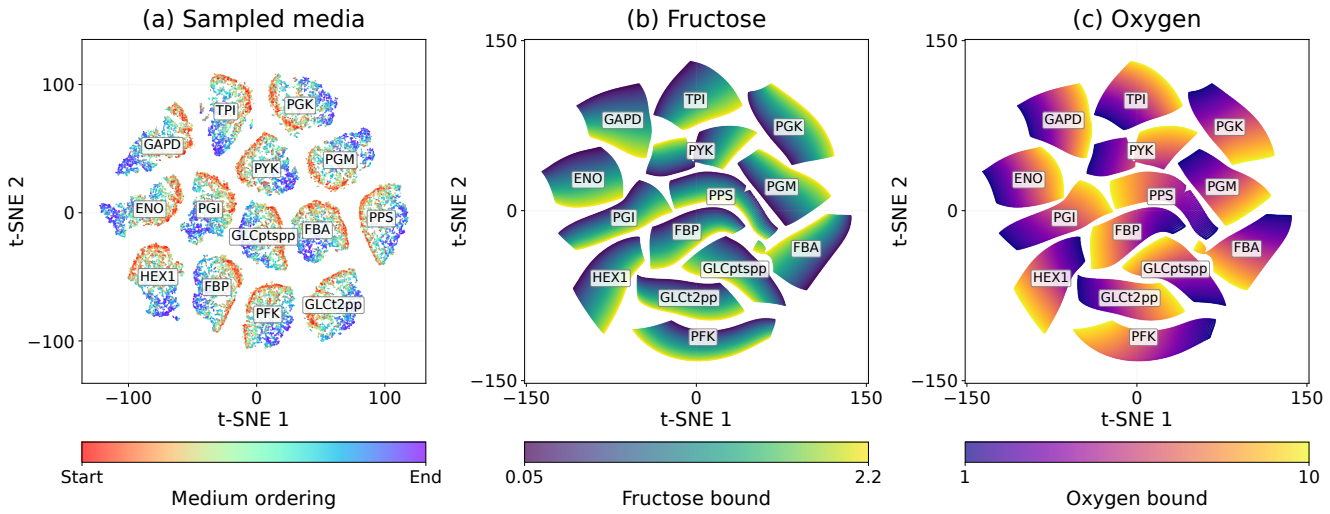


Figure 5. Nutrient-dependent reaction representations in model E. Each point represents one glycolysis-related reaction under one input condition. Left: 4,000 independently sampled test media, colored by position in a greedy nearest-neighbor ordering of standardized medium inputs. Middle and right: 10,000 controlled sweep conditions, colored by the fructose uptake bound (middle: 0.05–2.2) or oxygen uptake bound (right: 1–10), in mmol $\mathrm{gDW}^{-1}\,\mathrm{h}^{-1}$. The sweep panels share one t-SNE projection; the sampled-media panel uses a separate projection. Labels identify the reactions.

important when extending pretraining to a large range of nutrient conditions. A reaction that is inactive in one environment may well be active in another, and removing it would limit the metabolic responses the surrogate can learn.

Results across the different pretraining distributions show that FluxFormer performs well even when trained on nutrient conditions that differ from the experimental setting. Model E, which used a broad nutrient distribution without task-specific sampling, achieved nearly the same growth-prediction accuracy as the AMN-focused model A. Adding task-specific conditions in model D improved accuracy only slightly. This is encouraging for developing a more general FluxFormer, since training on a wider range of nutrients makes the model useful for different applications without requiring separate pretraining for each task.

The missing-flux experiment provides further evidence that FluxFormer can be used to support a range of different experimental tasks. By learning to infer additional exchange constraints from omics measurements, FluxFormer improved the subsequent pFBA predictions. In both applications, the pretrained surrogate was kept frozen, and a small neural network was trained to learn the mapping from the experimental inputs to the metabolic constraints. This allows the metabolic knowledge learned from the flux simulations to be reused even when the data covers only growth or a very limited set of fluxes. The nutrient-dependent reaction representations indicate that FluxFormer learns relationships between reactions that change with the metabolic conditions.

The biological realism of FluxFormer ultimately depends on the metabolic model and simulation assumptions used for pretraining. As with FBA, the simulated fluxes are governed by stoichiometric constraints, reaction bounds, and an assumed cellular objective[6]. Any representation of metabolic regulation or response to environmental stress is therefore only represented to the extent that it is included in the simulations. Future training could incorporate additional physiological constraints and experimental data in order to capture these effects more realistically.

An important next move toward a universal FluxFormer is joint pretraining across different microorganisms. This would involve a shared representation of the corresponding metabolic reactions while accounting for organism-specific differences in network structure and compartments. Training across several bacterial and fungal metabolic models would enable the surrogate to learn metabolic relationships that are common across species. One way to assess this is to test its predictions on organisms not in the pretraining set to see whether the learned relationships transfer to previously unseen metabolic networks. Gene knockouts and other physiological perturbations would be a natural extension of this to cover a wider range of conditions.

For industrial applications, the simulated nutrient conditions must ultimately reflect the actual composition of the raw material. Industrial and agricultural side streams contain complex nutrient profiles and may also contain inhibitory compounds which will impact microbial growth and production[38]. Literature-informed constraints and experimentally measured feedstock compositions could provide a more realistic set of training conditions. Environmental factors such as pH and temperature would also need to be included where appropriate data is available.

Our long-term goal is to develop a single FluxFormer that can predict metabolism across different microorganisms and nutrient conditions. One application is the conversion of industrial or agricultural side streams into microbial biomass and other

useful products[39]. Our model could be used to identify microbes that are capable of utilizing a given feedstock and to predict which growth conditions would be optimal for product formation. To assess its practical utility, we would need experimental measurements of a range of organisms grown on representative side streams. We could then test whether the organisms and cultivation conditions FluxFormer identifies as promising perform as predicted.

Conclusion

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Supplementary Material

Simulated pretraining distributions

All pretraining datasets were generated from the iML1515 reconstruction with biomass growth as the primary objective. pFBA was used with `fraction_of_optimum = 0.999`. Each dataset contained 1,000,000 accepted samples, and the full iML1515 reaction vector was retained as the prediction target. Unless otherwise stated, exchange-bound magnitudes are reported in mmol gDW^{-1} h^{-1}.

General sampling formulation. For a source-selection regime z, let $\mathscr{C}_z$ denote its pool of selectable organic sources. The number of selected sources was drawn as

$$k \sim p_z(k),$$

where p_z depends on the regime. Conditional on k, k distinct sources were selected uniformly without replacement,

$$\mathscr{C}_{\text{act}} \sim \text{Unif}\left(\{A \subseteq \mathscr{C}_z : |A| = k\}\right).$$

For each selectable exchange i, the uptake bound was then set to

$$\ell_i = \begin{cases} -r_i, & i \in \mathscr{C}_{\text{act}}, \\ 0, & i \notin \mathscr{C}_{\text{act}}, \end{cases} \qquad r_i \sim D_z,$$

where r_i is the sampled uptake-bound magnitude and D_z is the regime-specific sampling distribution. The negative sign follows the standard FBA convention for uptake.

For distributions containing several regimes, the regime was first selected according to the corresponding mixture probabilities. The source-selection procedure above was then applied within the selected regime. Fixed sources, such as glucose in B-style and mixed C conditions, were added separately. Basal exchanges and other variable exchanges, including oxygen and secretion bounds, were then set according to the selected regime before solving the condition with pFBA.

Common basal exchanges. The same 22 basal exchanges were used in all pretraining distributions. Their uptake bounds were fixed at 10. In B-style conditions, the carbon dioxide secretion bound was varied separately as described below.

Table 4. Common basal exchanges used in all pretraining distributions.

Group	Exchanges
Major inorganic nutrients	Ammonium, phosphate, sulfate, and carbon dioxide
Water and proton	Water and H^+
Metal and alkali ions	Fe^{3+}, Fe^{2+}, Mn^{2+}, Zn^{2+}, Mg^{2+}, Ca^{2+}, Ni^{2+}, Cu^{2+}, Co^{2+}, K^+, and Na^+
Other inorganic ions	Chloride, molybdate, tungstate, selenate, and selenite

Distribution A: AMN-specific conditions

Distribution A was based on the nutrient setting used in the AMN study. Ten carbon sources were included in the variable nutrient set: D-ribose, maltose, melibiose, trehalose, D-fructose, D-galactose, acetate, D-lactate, succinate, and pyruvate. For each sample, the number of selected sources, k, was drawn uniformly from $\{1,2,3,4\}$, after which k distinct sources were selected uniformly without replacement from the ten-source set. Their uptake bounds were sampled loguniformly from 0.05 to 10. Glycerol and four supplemented amino acids, L-alanine, L-proline, L-threonine, and glycine were present at fixed uptake bound of 2.2. Oxygen was sampled loguniformly from 1 to 25, while the 22 basal exchanges listed in Table 4 were fixed at 10.

Distribution A broadened the original AMN simulation setting in two ways. The AMN simulations used the carbon-source combinations represented in the 110 experimental media, whereas A sampled one to four sources independently from the ten-source set. In addition, oxygen was varied in A rather than kept fixed as in the AMN simulations.

Distribution B: MINN-specific conditions

Distribution B was based on the glucose-minimal setting used in MINN. Glucose uptake caps were sampled from a continuos uniform distribution over 1–15, while oxygen uptake caps were sampled loguniformly from 1–25. Carbon dioxide, ethanol, and acetate secretion bounds were sampled from continuous uniform distributions over 0–20, 0–2, and 0–8, respectively. These

ranges were chosen to cover the variability represented in the Ishii data used in the MINN experiments. The remaining basal exchanges were fixed at 10.

Distribution A ∪ B: literal mixture of the task-specific distributions

Distribution A ∪ B was formed as an exactly balanced mixture of A and B distributions. For one million accepted samples, 500,000 samples were generated from A and 500,000 from B. A 40-exchange input vocabulary contained the union of the exchanges required by both regimes. Each sample followed either the complete A or the complete B sampling procedure, and no mixed conditions were generated.

Distribution C: task-relevant bridge distribution

Distribution C combined A-like, B-like, and mixed conditions, selected independently with probabilities 0.40, 0.40, and 0.20, respectively. A-like conditions followed Distribution A's source-selection procedure, with independently sampled loguniform uptake caps of 0.05–10 for selected carbon sources and 1–25 for oxygen. Glycerol and the four supplemented amino acids had fixed uptake caps of 2.2.

B-like conditions used glucose uptake caps of 1–15 sampled from a continuos uniform distribution and loguniform oxygen uptake caps of 1–25. Carbon dioxide secretion caps were sampled from a continuous uniform distribution over 0–20. Ethanol and acetate secretion caps were sampled from continuous uniform distributions over 0–2 and 0–8, respectively. All five caps were sampled independently.

Mixed conditions began with this same B-like medium. The number of additional A carbon sources was drawn uniformly from $\{1,2,3\}$, giving each count probability $1/3$. Conditional on this count, distinct sources were selected uniformly without replacement from the nine A carbon sources excluding acetate. Acetate retained its B-like secretion-only role. Each added source received an independent loguniform uptake cap of 0.05–10. Glycerol and amino-acid supplements were absent from B-like and mixed conditions.

The 21 common basal exchanges other than carbon dioxide had fixed uptake caps of 10 in every regime. A-like conditions additionally allowed carbon dioxide uptake at 10; B-like and mixed conditions instead used the sampled carbon dioxide secretion cap. Continuous cap draws were rounded to two decimal places. Distribution C used a 40-exchange input vocabulary. Conditions were solved using pFBA. The 0.40/0.40/0.20 mixture probabilities were used when sampling new conditions. Conditions without an optimal pFBA solution were discarded and resampled, so the final accepted dataset was not required to contain these three regimes in exactly the same proportions.

Literature-guided construction of the broad D/E source pool

We constructed the broad source pool for Distributions D and E in two stages. First, we identified candidate organic sources computationally from iML1515 using a COBRApy-based exploration notebook[40]. The notebook screened organic exchange reactions by allowing uptake of each candidate and solving the model for biomass production. Exchanges that supported a positive biomass flux under the screening conditions were retained as candidate sources and saved together with their exchange identifiers and predicted biomass fluxes. This produced 185 candidate exchanges for the subsequent literature review.

We then evaluated the candidate set using an LLM-assisted review of the literature. The review focused on unmodified *E. coli* k-12 MG1655 and on organic compounds occurring in industrial and agricultural side streams, complex cultivation media, and other potentially relevant nutreint environments. By design, we retained all 15 organic inputs used in the AMN setting and the glucose input used in the MINN setting. The remaining sources were selected through the literature review.

For each candidate, three types of evidence were considered separately: whether the compound had been identified in a relevant feedstock or cultivation medium, whether microbial consumption of the combound had been measured, and whether unmodified *E. coli* could consume it or grow using it. The review also considered strain dependence, genetic engineering or adaptation requirements, stereochemistry, processing conditions, and whether a compound was measured directly or appeared after hydrolysis or fermentation. Primary studies reporting quantified composition or direct cultivation measurements were preferred.

Each candidate was assigned an inclusion or exclusion decision based on the iML1515 screening result and the literature evidence. The final pool contained 48 selectable organic sources: the 16 task-related sources retained by design and 32 additional sources selected through the literature review. The complete pool is listed in Table 5. The literature review also retained the evidence and rationale for excluded candidates so that the final pool could be traced back to the original iML1515 candidate set.

The maximum number of sources selected in a broad condition was determined as part of the same review. Measured compositions from several feedstock and cultivation-medium families were examined to estimate the number of relevant organic substrates occurring together under comparable inclusion criteria. These observations were considered together with the one-million-sample simulation budget and the rapid increase in the number of possible source combinations as the mixture size increases. The quantified cross-family compositions generally contained between two and nine eligible sources under the

Table 5. Selectable organic sources used in the common D/E broad nutrient pool.

Group	Compounds
Sugars, sugar alcohols, and related substrates	N-acetyl-D-glucosamine, glycerol, D-mannose, D-ribose, D-sorbitol, D-glucose, L-arabinose, D-fructose, D-galactose, D-xylose, melibiose, trehalose, L-fucose, maltopentaose, L-rhamnose, lactose, maltotriose, D-mannitol, maltose, D-glucosamine, and maltotetraose
Organic acids and related substrates	L-lactate, D-gluconate, acetate, 2-oxoglutarate, pyruvate, succinate, fumarate, D-lactate, glycolate, D-galacturonate, propionate, and L-malate
Amino acids	L-glutamine, L-proline, L-alanine, L-arginine, L-aspartate, L-glutamate, glycine, L-serine, L-tryptophan, L-asparagine, and L-threonine
Nucleosides	cytidine, adenosine, inosine, and uridine

adopted criteria. A maximum of $K = 10$ therefore provided some additional headroom while limiting the high-cardinality part of the combinatorial sampling space. This value was treated as a practical coverage limit rather than as a biological maximum.

The complete literature-review prompt, candidate-level inclusion and exclusion decisions, and supporting primary-literature records were retained with the accompanying study materials.

Distributions D and E: broad nutrient conditions

Distributions D and E used the common 48-source pool defined above. The broad-condition sampling procedure shared by D and E is denoted by P_G. Uptake of all unselected organic sources was closed before each sample was generated. Between one and ten distinct sources were then selected from the 48-source pool.

The number of selected sources, k, followed the truncated geometric distribution

$$P(k) = \frac{(2/3)^{k-1}}{\sum_{j=1}^{10}(2/3)^{j-1}}, \qquad k = 1, \ldots, 10.$$

This distribution has an expected value of approximately 2.82 sources per condition. It gives more weight to smaller mixtures, whose combination space can be sampled more densely. Larger mixtures remain represented, but occur less frequently.

Conditional on k, the source identities were selected uniformly without replacement. Uptake bounds for the selected sources were sampled independently from a log-uniform distribution between 0.05 and 10. Oxygen was sampled independently from a log-uniform distribution between 1 and 25 and was not counted among the k selected organic sources. The common basal exchanges listed in Table 4 were fixed at uptake bounds of 10. Carbon dioxide retained its model secretion upper bound in the broad conditions. The selected sources were made available for uptake, but pFBA could assign zero flux to some of them.

Distribution E consisted entirely of conditions sampled from P_G. Distribution D used P_G for 80% of its conditions, while 10% followed the A sampling procedure and 10% followed the B sampling procedure. For a dataset of one million samples, this corresponded to 800,000 broad conditions, 100,000 A-style conditions, and 100,000 B-style conditions. The additional task-specific samples increased the sampling density around the two downstream experimental condition spaces while retaining the same broad condition space as Distribution E.